

ABS

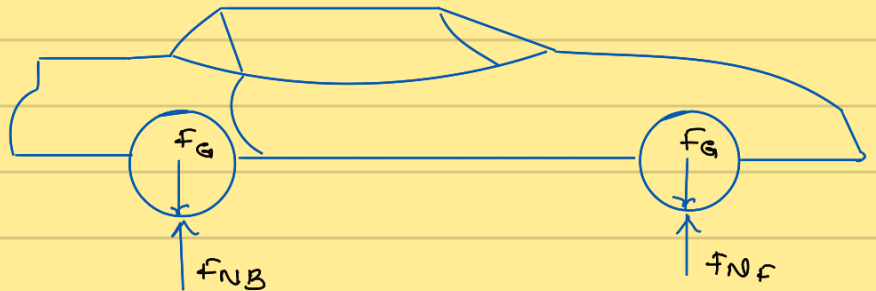
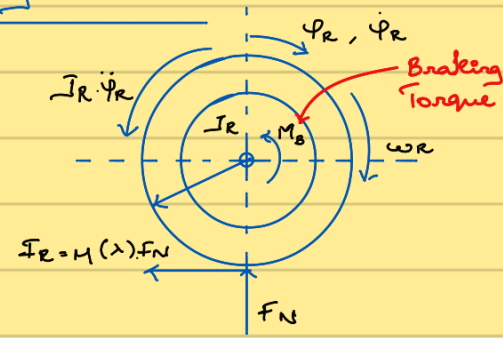
Equations of Motion for a single wheel

Torque equilibrium at the center of the wheel $\Sigma M = 0$,
 M_R at the contact patch is

$$M_R = r_R \cdot F_R$$

$$J_R \cdot \ddot{\varphi}_R + M_B - M_R = 0$$

$$\ddot{\varphi}_R = \frac{F_R \cdot r_R - M_B}{J_R} \quad \text{--- (1)}$$



For normal cases,

$$F_{NB} = F_G \quad \& \quad F_{NF} = F_G$$

While braking, pitching happens and weight is transferred to the front,

$$\Rightarrow F_{NB} = 0.5 F_G \quad \& \quad F_{NF} = 1.5 F_G$$

$$\therefore \text{For a front wheel } F_N = 1.5 F_G = 1.5 mg$$

$$F_R = \mu \cdot F_N = 1.5 \mu mg //$$

Slippage is given by $\lambda = \frac{v_f - v_R}{v_f}$, $v_R = r_R \cdot \omega_R$

$$\mu(\lambda) = c_1 (1 - e^{-c_2 \lambda}) - c_3 \lambda \quad \text{--- (2)}$$

$$\mu_{\max} \text{ at } \lambda = 0.13$$

Equations of Motion for the whole car

Frictional force $\rightarrow F_R$

$$\text{Aerodynamic drag} \rightarrow F_L = \frac{c_w A \rho v_f^2}{2}$$

$$\sum F = 0 \quad \Rightarrow \quad m \ddot{x}_F = -F_R - F_L$$

$\Rightarrow$

$$\ddot{x}_F = \frac{-F_R - F_L}{m}$$

$\Rightarrow$

$$\ddot{x}_F = \frac{1}{m} \left(-H(\omega) F_N - \frac{c \omega A P v_F^2}{2} \right) \quad \text{--- (3)}$$